الأسئلة المقالية — 10 عام · مذاكرة الهيكل الاختبار النهائي للفصل الاول

الاسم: _____ المدرسة: _____ التاريخ: _____ (21 سؤالاً)

LO21 جمع كثيرات الحدود وطرحها · Add and subtract polynomials

21.1 LO21 · Reveal p. 636 · 27

Find the sum or difference.

$$(3c^3 - c + 11) - (c^2 + 2c + 8)$$

a. Show your simplification. (10)

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21.2 LO21 · Reveal p. 636 · 28

Find the sum or difference.

$$(z^2 + z) + (z^2 - 11)$$

a. Show your simplification. (10)

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21.3 LO21 · Reveal p. 636 · 29

Find the sum or difference.

$$(2x - 2y + 1) - (3y + 4x)$$

a. Show your simplification. (10)

21.4 LO21 · Reveal p. 636 · 30

Find the sum or difference.

$$(4a - 5b^2 + 3) + (6 - 2a + 3b^2)$$

a. Show your simplification. (10)

22.1 LO22 · Reveal p. 650 · 18

Find the product.

$$(2y - 11)(y^2 - 3y + 2)$$

a. Show all partial products and combine like terms. (10)

22.2 LO22 · Reveal p. 650 · 19

Find the product.

$$(4a + 7)(9a^2 + 2a - 7)$$

a. Show all partial products and combine like terms. (10)

22.3 LO22 · Reveal p. 650 · 20

Find the product.

$$(m^2 - 5m + 4)(m^2 + 7m - 3)$$

a. Show all partial products and combine like terms. (10)

22.4 LO22 · Reveal p. 650 · 21

Find the product.

$$(x^2 + 5x - 1)(5x^2 - 6x + 1)$$

a. Show all partial products and combine like terms. (10)

23.1 LO23 · Reveal p. 576 · 25

A square field has side length $5x^3y^5$. What is its area in terms of x and y?

a. Show the area calculation. (10)

23.2 LO23 · Reveal p. 576 · 27

A cube-shaped shipping box has side length $4m^8n^4$. What is its volume in terms of m and n?



a. Show the volume calculation. (10)

23.3 LO23 · Reveal p. 578 · 56

Consider the two expressions.

$$(m^p)(m^q) \quad \text{and} \quad (m^p)^q$$

a. What must be true about p and q for the expressions to be equal? Give an example. (5)

b. Assuming m, p and q are positive integers, can $(m^p)(m^q) > (m^p)^q$ ever be true? Explain. (5)

23.4 LO23 · Reveal p. 578 · 57

For nonzero real a and b and integers m and t, simplify and explain.

$$\left(-\frac{a^m}{b^t}\right)^{2t}$$

a. Apply the power rules and explain the sign. (10)

23.5 LO23 · Reveal p. 578 · 58

Consider $y=x$, $y=x^2$ and $y=x^3$.

a. For each equation, write the related expression and record the power of x. (3)

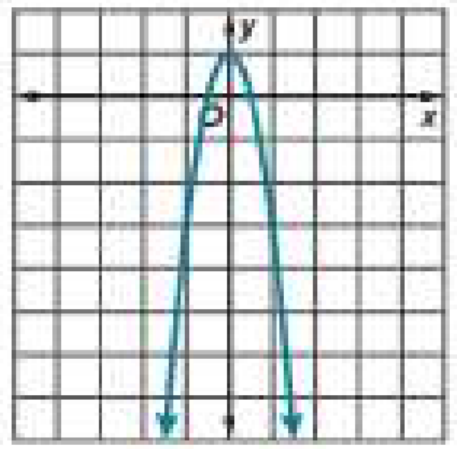
b. Graph each equation using a graphing calculator. (3)

c. Classify each graph as linear or nonlinear. (1)

d. Explain how to determine whether an equation, or its related expression, is linear or nonlinear without graphing. (3)

24.1 LO24 · Reveal p. 695 · 1

Identify the axis of symmetry, vertex and y-intercept of the graph. Then describe the end behavior.

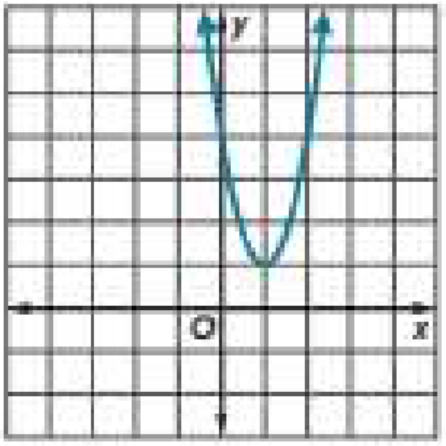


a. Axis of symmetry and vertex. (5)

b. y-intercept and end behavior. (5)

24.2 LO24 · Reveal p. 695 · 2

Identify the axis of symmetry, vertex and y-intercept of the graph. Then describe the end behavior.

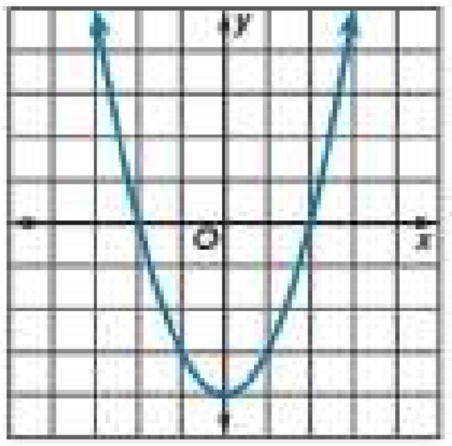


a. Axis of symmetry and vertex. (5)

b. y-intercept and end behavior. (5)

24.3 LO24 · Reveal p. 695 · 3

Identify the axis of symmetry, vertex and y-intercept of the graph. Then describe the end behavior.

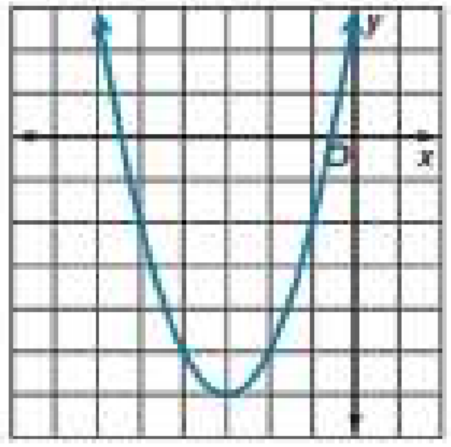


a. Axis of symmetry and vertex. (5)

b. y-intercept and end behavior. (5)

24.4 LO24 · Reveal p. 695 · 4

Identify the axis of symmetry, vertex and y-intercept of the graph. Then describe the end behavior.



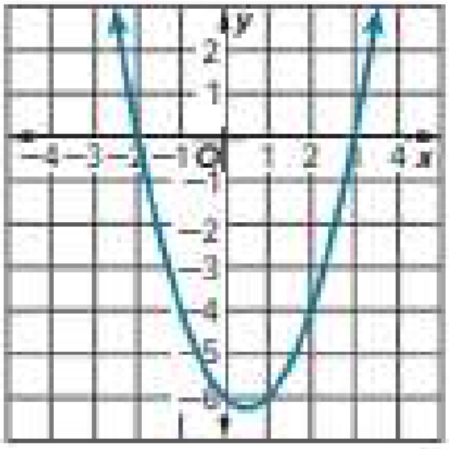
a. Axis of symmetry and vertex. (5)

b. y-intercept and end behavior. (5)

LO25 كتابة الدالة التربيعية من الرسم · Write a quadratic from its graph

25.1 LO25 · Reveal p. 722 · 31

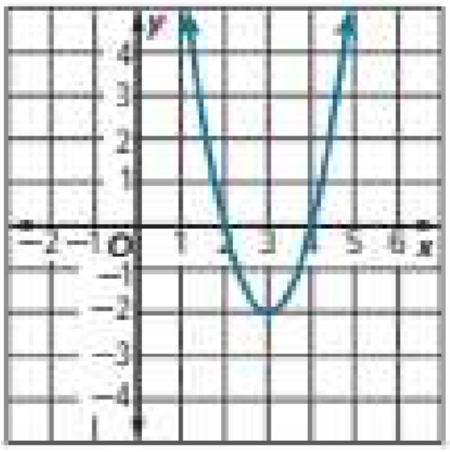
Write a quadratic function for the given graph. Give the equation in standard form.



a. Show how you find the equation. (10)

25.2 LO25 · Reveal p. 722 · 32

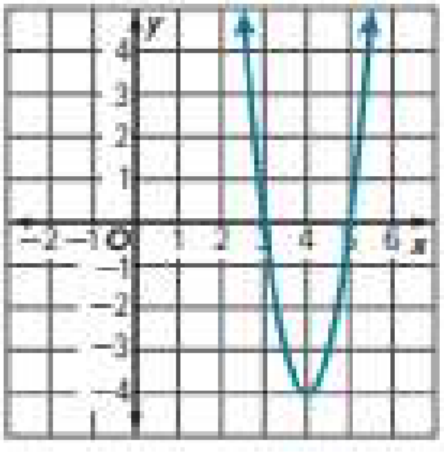
Write a quadratic function for the given graph. Give the equation in standard form.



a. Show how you find the equation. (10)

25.3 LO25 · Reveal p. 722 · 33

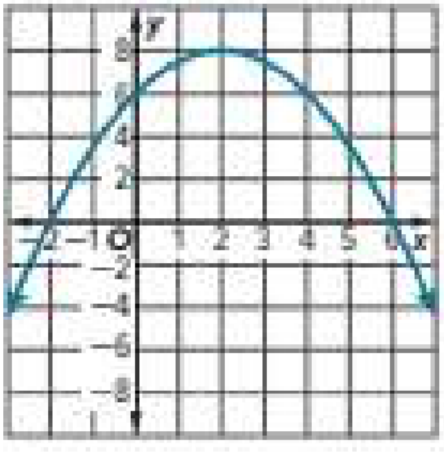
Write a quadratic function for the given graph. Give the equation in standard form.



a. Show how you find the equation. (10)

25.4 LO25 · Reveal p. 722 · 34

Write a quadratic function for the given graph. Give the equation in standard form.



a. Show how you find the equation. (10)